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Global-ocean redox variation during the middle-late Permian through Early Triassic based on uranium isotope and Th/U trends of marine carbonates

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ABSTRACT

Uranium isotopes ($^{238}U/^{235}U$) in carbonates, a proxy for global-ocean redox conditions owing to their redox sensitivity and long residence time in seawater, exhibit substantial variability in the Daxiakou section of south China from the upper-middle Permian through the mid-lower Triassic (~9 m.y.). Middle and late Permian ocean redox conditions were similar to that of the modern ocean and were characterized by improving oxygenation in the ~2 m.y. prior to the latest Permian mass extinction (LPME), countering earlier interpretations of sustained or gradually expanding anoxia during this interval. The LPME coincided with an abrupt negative shift of >0.5‰ in $\delta^{238}U$ that signifies a rapid expansion of oceanic anoxia. Intensely anoxic conditions persisted for at least ~700 k.y. (Griesbachian), lessening somewhat during the Dienerian. Th/U concentration ratios vary inversely with $\delta^{238}U$ during the Early Triassic, with higher ratios reflecting reduced U concentrations in global seawater as a consequence of large-scale removal to anoxic facies. Modeling suggests that 70%–100% of marine U was removed to anoxic sinks during the Early Triassic, resulting in seawater U concentrations of <5% that of the modern ocean. Rapid intensification of anoxia concurrent with the LPME implies that ocean redox changes played an important role in the largest mass extinction event in Earth history.

INTRODUCTION

The latest Permian mass extinction (LPME) was the largest biocrisis of the Phanerozoic, resulting in a loss of >90% of species globally and the nearly total restructuring of marine and terrestrial ecosystems (Chen and Benton, 2012). A broad consensus has emerged that the LPME was triggered by the massive eruptions of the Siberian Traps through injection of large quantities of CO_2 and CH_4 to the atmosphere, leading to extreme global warming and widespread marine anoxia (Korte and Kozur, 2010; Payne and Clapham, 2012). However, the timing of onset, the duration, and the extent of oceanic anoxia are debated, leaving the causal relationship between deoxygenation and the LPME uncertain. Arguments have been made for widespread anoxia starting as early as 10 m.y. prior to the LPME (Isozaki, 1997; Kato et al., 2002; Wignall and Twitchett, 2002), although more recent studies have concluded that ocean redox conditions were spatially heterogeneous (Bond and Wignall, 2010) and that anoxia was confined mainly to intermediate-depth oceanic oxygen-minimum zones (OMZs; Algeo et al., 2010, 2011; Winguth and Winguth, 2012). However, all of the proxy-based studies to date have relied on proxies that evaluate local rather than global water mass redox conditions.

The ratio of $^{238}U/^{235}U$ (expressed as $\delta^{238}U$ in units of per mil [‰] deviation from a standard) is a recently developed proxy for globally integrated paleocean redox conditions (Weyer et al., 2008; Romaniello et al., 2013; Dahl et al., 2014). It was applied to the Permian–Triassic interval by Brennecka et al. (2011) and Lau et al. (2016); both studies documented an abrupt ~0.3‰ negative $\delta^{238}U$ shift at the LPME in sections in south China and Turkey, corresponding to an ~6× increase in the global area of anoxic sedimentation. The Brennecka et al. (2011) study examined only a narrow, ~200-k.y.-long interval around the LPME, whereas the Lau et al. (2016) study reported U-based ocean redox trends for an ~20-m.y.-long interval spanning the latest Permian through Middle Triassic. To date, U isotopes in marine carbonates have not been examined for the full upper Permian to test ideas concerning global ocean redox changes leading into the LPME event. We present $\delta^{238}U$ and Th/U trends from upper-middle Permian through lower Triassic marine carbonates at the Daxiakou section in south China (Hubei Province) to evaluate global seawater redox conditions in an ~8-m.y.-long interval prior to the LPME and <1 m.y. after the event.

BACKGROUND

Geologic Setting

Daxiakou is located in Hubei Province in south-central China, an area close to the paleoequator during the late Permian to Early Triassic (Fig. DR1 in the GSA Data Repository[1]). The study section was deposited on a carbonate ramp, at water depths of >200 m, on the northern (paleowestern) margin of the Yangtze platform (Zhao et al., 2005). The section comprises the middle and upper Permian Maokou, Wujiaping, and Dalong Formations and the lower Triassic Daye Formation (Fig. 1) that consist of limestones and argillaceous limestones (Table DR1). Precise age control is provided by the dozens of ash layers near the Permian-Triassic boundary, conodont biostratigraphy, and carbonate $\delta^{13}C$ stratigraphy (Wang and Xia, 2004; Tong et al., 2007; Shen et al., 2012; Zhao et al., 2013).

U Isotope and Th/U Systematics

The main source for U to the global ocean is weathering of continental crust, and the main sinks are suboxic and anoxic continental margin

[1]GSA Data Repository item 2017041, supplementary information, figures, and data tables, is available online at www.geosociety.org/pubs/ft2017.htm, or on request from editing@geosociety.org.

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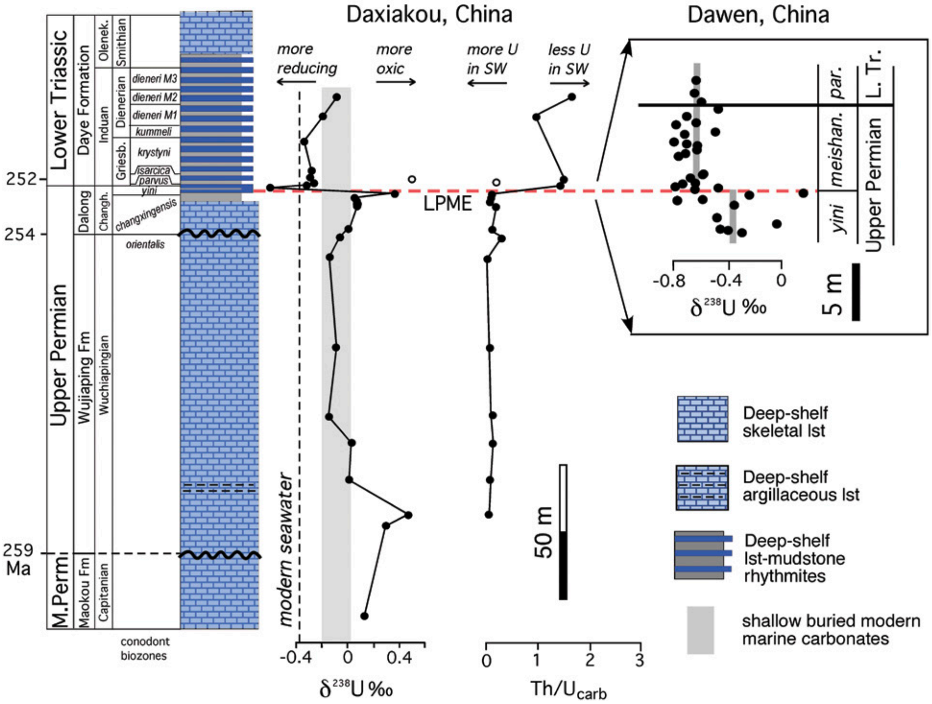

Figure 1. Generalized lithologic column, conodont biostratigraphy (Wang and Xia, 2004; Zhao et al., 2005, 2013), U isotopes, and Th/U ratios for the carbonate fraction, Daxiakou section, south China. Open circles are interpreted as outliers. LPME—latest Permian mass extinction; lst—limestone. Upper right: δ^{238}U profile from Dawen, Guizhou Province, south China (Brennecka et al., 2011) that spans only an ~200 k.y. interval straddling the LPME. Note the similarities in timing and magnitude of δ^{238}U shifts between these two sections, which are separated by >500 km, arguing for a global signal.

sediments, altered ocean crust, and biogenic carbonates (Morford and Emerson, 1999; Tissot and Dauphas, 2015). In oxidizing marine waters, uranium is present mainly as U(VI) in the large uranyl ion (UO_2^{2+}), which forms soluble carbonate and phosphate complexes (Weyer et al., 2008). In reducing environments, reduction to U(IV) leads to formation of relatively insoluble hydroxide and phosphate complexes and organo-metallic ligands (Klinkhammer and Palmer, 1991). The two most abundant isotopes of U are ^{238}U (~99.2%) and ^{235}U (~0.7%). Fractionation occurs in low-temperature aqueous systems through reduction of U(VI) to U(IV), with microbially mediated enrichment of ^{238}U into U(IV) (Weyer et al., 2008; Basu et al., 2014; Stylo et al., 2015). This chemical behavior, along with the long residence time of U in seawater (~400 k.y.; Ku et al., 1977), permits use of ^{238}U/^{235}U ratios as a tracer of global marine redox history. During times of expanding oceanic anoxia, increased sequestration of U(IV) in anoxic deposits results in ^{238}U depletion of seawater, and thus lower ^{238}U/^{235}U ratios in marine calcite. The ratio is defined as $[(^{238}U/^{235}U)_{sample}/(^{238}U/^{235}U)_{standard} - 1] \times 1000‰$, where the standard is NBL-112A (also CRM-112A), which has a ^{238}U/^{235}U ratio of 137.832 ± 0.026 (Cheng et al., 2013).

Th/U ratios can be used to independently evaluate changing oceanic redox conditions (cf. Wignall and Myers, 1988). Th is sourced from continental weathering and has only one oxidation state (IV); therefore, its concentration in seawater is not affected by water mass redox changes. Because U is sensitive to redox conditions and sequestered in anoxic facies, an increase in the Th/U ratios of marine carbonates represents enhanced rates of U removal to the sediment linked, presumably, to expanded areas of reducing facies. (See the Data Repository for discussion of other factors influencing Th/U ratios.)

RESULTS

The study section yielded δ^{238}U ranging from –0.60‰ to +0.47‰ (Fig. 1; Table DR1). Values were relatively high (>0‰) during the late-middle through early-late Permian before declining to nearly uniform values of –0.14‰ ± 0.04‰ through most of the Wuchiapingian. The uppermost Permian is marked by a trend toward higher values, ~+0.10‰ in the upper *changxingensis* conodont zone, followed by a single-point positive excursion to +0.36‰ in the *yini* Zone. At the LPME horizon, δ^{238}U shifts abruptly to a single-point minimum of –0.60‰. From the base of the Triassic (basal *parvus* Zone) through the upper Griesbachian (upper *krystyni* Zone), δ^{238}U is nearly uniform at –0.30‰ ± 0.03‰. The overlying Dienerian is characterized by a weak trend toward higher values, culminating at –0.08‰ (Fig. 1). We validated these results by multiple δ^{238}U measurements of modern *Porites* coral and the IAPSO (International Association for the Physical Sciences of the Oceans) seawater standard, obtaining values within analytical error of previously reported values (see the Data Repository).

The Th/U ratios for the carbonate fraction (carb) range from 0.021 to 1.6 (Fig. 1; Table DR1). Th/U_{carb} shows uniformly low values (<0.25) throughout the upper Permian up to the LPME horizon, which is marked by an abrupt shift to 1.4, and most lower Triassic samples yield ratios >1.0. Th/U ratios for whole-rock (WR) analysis range from 0.12 to 7.4. Although Th/U_{WR} values are generally larger than Th/U_{carb} values, the two profiles exhibit similar trends (Figs. DR3 and DR4).

DISCUSSION

Several lines of evidence indicate that the measured δ^{238}U and Th/U profiles record primary global seawater trends. First, the trends through the immediate LPME interval are similar in timing and magnitude to those reported for coeval sections in south China (Brennecka et al., 2011; Lau et al., 2016), Turkey (Lau et al., 2016), and Oman (Ehrenberg et al., 2008). Second, δ^{238}U and Th/U values for the lower Triassic covary negatively [$r = -0.71$, $p(\alpha) < 0.001$], and δ^{238}U does not exhibit significant covariation with total organic carbon, insoluble residue content, U_{WR}, or Th/U_{carb} values (Fig. DR4). Third, the Daxiakou trends do not appear to be controlled by lithofacies; uppermost Permian shaly limestones yield δ^{238}U and Th/U values similar to those of the underlying clean limestones (Fig. 1). Fourth, local redox conditions at Daxiakou, as determined by Fe-speciation and pyrite S isotope analyses (Shen et al., 2016), are largely unrelated to the global redox pattern inferred from U isotope trends.

Quantitative assessment of Permian–Triassic redox variations requires compensating for early diagenetic effects on carbonate δ^{238}U. Under anoxic or sulfidic pore-water conditions, precipitation of ^{238}U-enriched secondary carbonate cement yields bulk-carbonate δ^{238}U higher than that of the primary marine components. The magnitude of this effect in modern carbonates was assessed by Romaniello et al. (2013); a 0.2‰–0.4‰ offset between carbonate bioclasts and bulk carbonate (including secondary cements) was found in samples from <40 cm below the seafloor. In ancient carbonates, this effect can be removed by subtracting a diagenetic-correction factor (Δdiag) from measured δ^{238}U values (see the Data Repository for a discussion of the use of a 0.5‰ Δdiag). The Δdiag-corrected δ^{238}U profile for Daxiakou suggests that, relative to the modern ocean, late-middle and early-late Permian oceans were similar, middle- and latest Permian oceans somewhat more reducing, and Early Triassic oceans substantially more reducing.

The magnitude and significance of the end-Permian negative $\delta^{238}U$ excursion were considered by Brennecka et al. (2011) in their study of the Dawen section, located >500 km southwest of Daxiakou (Fig. DR1). Brennecka et al. (2011) documented a –0.30‰ shift across the LPME (Fig. 1) that they equated with an ~6× expansion of anoxic facies (i.e., from 10% to 60% of the U sink flux) based on a mass-balance model of the seawater U cycle. Modeling of U sink fluxes using our $\delta^{238}U$ profile (Fig. DR5) suggests that the anoxic sink flux expanded from 0%–40% in the pre-LPME interval to 70%–100% in the latest Permian and Griesbachian, before declining to <50% by the Dienerian (Fig. 2). Given that anoxic facies compose just ~0.2% of the total modern seafloor area (Tissot and Dauphas, 2015), these redox changes, although significant, nonetheless imply only modest areas of anoxic deposition (~2%–3% of total ocean floor) rather than whole-ocean anoxia (e.g., Isozaki, 1997; Wignall and Twitchett, 2002). A change of this magnitude is consistent with the interpretations of studies invoking development and/or expansion of euxinia within end-Permian oceanic OMZs, with the deep seafloor remaining largely suboxic (Algeo et al., 2010, 2011; Winguth and Winguth, 2012). Because suboxic facies are associated with only a small fractionation of U isotopes (Weyer et al., 2008), the widespread existence of suboxic conditions in the global ocean would have had, at most, a limited effect on seawater $\delta^{238}U$ values. As with previous $\delta^{238}U$ studies (Brennecka et al., 2011; Dahl et al., 2014; Lau et al., 2016), we recognize an offset in timing between $\delta^{13}C$ (Fig. DR6) and $\delta^{238}U$ excursions (within the sampling resolution of ~2 m) that is not well understood. This apparent offset implies that the processes influencing global carbon burial rates were temporally and perhaps spatially distinct from those influencing global marine anoxia (e.g., Dahl et al., 2014).

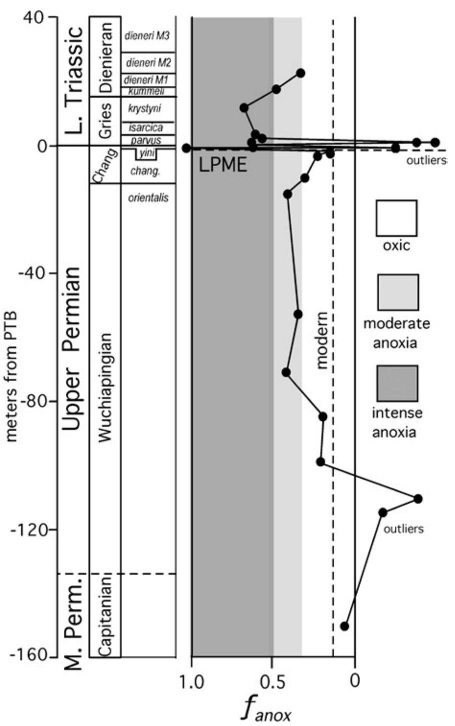

Figure 2. The fractional flux of seawater U to anoxic sinks (f_{anox}) through the late Permian and Early Triassic. Note the very low values of f_{anox} (oxic) in the <1 m.y. interval immediately preceding the latest Permian mass extinction (LPME) (horizontal dashed line) and the rapid shift to f_{anox} of 0.7–1.0 (anoxic) at the LPME. See the Data Repository (see footnote 1) for details of marine U-cycle modeling.

The duration of the negative $\delta^{238}U$ excursion at the LPME can be estimated from average sedimentation rates for the uppermost Permian at Daxiakou (Fig. DR2). The excursion occurs over <2.1 m of section, representing an interval of ≤370 k.y., slightly shorter than the modern seawater residence time of U. In order for the U isotopic composition of seawater to have changed so rapidly, the concentration of U in global seawater during the LPME event must have been considerably lower than at present. Removal of 70%–100% of seawater U to anoxic sinks equates with an Early Triassic seawater U concentration of <5% that of the modern ocean (Andersen et al., 2014). This inference is consistent with the observed abrupt increases in Th/U ratios at the LPME of the Daxiakou and Dawen sections. Globally, shallow-marine and deep-sea sections exhibit a significant decline in the burial flux of U at the LPME (Ehrenberg et al., 2008; Algeo et al., 2011), whereas upwelling depositional systems on the northwestern Pangean margin show an ~2–4× increase (Schoepfer et al., 2013). These contrasting patterns are consistent with a global decline in seawater U concentrations and a concurrent increase in the U burial flux on continental margins on which OMZs expanded.

The pattern of secular variation in ocean redox conditions during the late Permian and Early Triassic has been debated at length. Of particular interest is that our results do not support findings that suggest persistent anoxia throughout the late Permian (Isozaki, 1997; Kato et al., 2002; Wignall and Twitchett, 2002) or a gradual decline in dissolved oxygen that contributed to or foreshadowed the LPME (Cao et al., 2009; Song et al., 2012; Dustira et al., 2013). Our data argue for increasing oceanic oxygenation during the Changhsingian until an abrupt onset of intense anoxia at the LPME horizon. These results also do not support the recent hypothesis that low O_2 levels during the late Permian enhanced preservation of huge stores of degradable organic carbon to be utilized by newly evolved methanogens responsible for CH_4 bursts at the LPME (Rothman et al., 2014). These apparent inconsistencies between various redox proxies at different locations probably reflect inherent differences between global versus local redox signals, underscoring the unreliability of local proxies for documentation of integrated ocean redox trends (Fig. DR7). In addition, some studies have suggested that marine redox conditions ameliorated at least locally during the Griesbachian (Twitchett et al., 2004; Bond and Wignall, 2010). In contrast, our results, along with those of Lau et al. (2016), indicate that average global-ocean redox conditions remained strongly reducing through the mid-Dienerian and beyond.

CONCLUSIONS

U isotope data from upper-middle Permian through lower Triassic marine carbonates at Daxiakou, south China, provide a globally integrated record of oceanic redox variation for an ~8-m.y.-long interval preceding, and an ~700-k.y.-long interval following, the LPME. $\delta^{238}U$ trends suggest that, in contrast to previous studies, the oxidation state of the pre-LPME late Permian ocean was similar to that of the modern ocean and, in particular, that the >1 m.y. interval immediately preceding the LPME was characterized by oxic conditions. A rapid (≤370 k.y.) large (>0.5‰) shift toward more negative $\delta^{238}U$ at the LPME horizon marks a major expansion of oceanic anoxia that persisted for at least ~700 k.y. into the Early Triassic. Modeling suggests that 70%–100% of marine U was removed to anoxic sinks during the Early Triassic, resulting in seawater U concentrations of <5% that of the modern ocean. Rapid intensification of anoxia occurring at the same time as the largest mass extinction event in history suggests that ocean redox changes were an important factor in the LPME.

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